Mathematical Simulation of Blood Flow

A Literature Review and Idealized Stenosis Study

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June 16, 2026

Abstract.

This report reviews the mathematical and numerical setting for simulating idealized stenotic blood flow and frames a controlled reduced-order stenosis study. The literature-review portion fixes the continuum, constitutive, boundary-data, and model-hierarchy vocabulary needed to move from three-dimensional Navier–Stokes and generalized-Newtonian models to reduced 0D/1D/2D hemodynamic formulations. Pressure-ratio quantities are treated as computed model outputs unless an external clinical measurement comparison is explicitly introduced.

Keywords:

computational hemodynamics, blood flow simulation, stenotic vessels, Navier–Stokes, reduced-order models, idealized stenosis

Contents

1 Introduction and Report Scope

Coronary and arterial stenoses motivate a basic computational question: how do geometry, wall properties, rheology, and boundary data determine the state of blood flow through an idealized narrowed vessel? Clinical pressure-ratio measurements motivate the question because anatomical narrowing alone does not determine functional obstruction. This report, however, is a mathematical and numerical study, not a clinical validation study.

The report fixes the vocabulary needed to move from continuum assumptions, to the three-dimensional incompressible Navier–Stokes and generalized-Newtonian settings, to reduced 0D/1D/2D hemodynamic model classes, and finally to a selected one-dimensional numerical study of an idealized stenosis. Its contribution is twofold: it organizes the relevant mathematical and numerical literature, and it frames a controlled computation for a smooth idealized stenosed vessel.

Add figure showing vessel

Pressure-ratio quantities in this report are computed model outputs, not clinical FFR or CT-FFR measurements, unless an external measurement comparison is explicitly introduced. Convention C.2 in Appendix C records the pressure-reference, gauge-pressure, high-flow-window, and admissibility conventions used for any reduced pressure-ratio output.

Literature Review Roadmap

1. Section 1.1 fixes the continuum, kinematic, balance-law, and constitutive vocabulary used later in the report.
2. Section 1.2 identifies the governing equations, boundary-data vocabulary, and model hierarchy.
3. Later literature-review sections use this vocabulary to compare classical finite-difference, finite-volume, and finite-element approaches for hemodynamic models.
4. The numerical-study portion then applies the selected reduced model to a controlled idealized stenosis computation.

1.1 Continuum and Constitutive Assumptions

This section moves from the macroscopic scale assumption, to continuum field variables, to balance-law vocabulary, and finally to Newtonian versus generalized-Newtonian stress closures.

1.1.1 Physiological Scale Boundary

The present work is posed at the macrocirculatory arterial scale. At this scale, blood is commonly modeled as a viscous fluid whose bulk behavior can be described by continuum mechanics. The goal is not to resolve the effects of individual blood cells, platelets, or plasma constituents but rather to predict macroscopic observables such as pressure, flow rate, pressure drop, and pressure ratio. [1, Sec. 1.1.2, pp. 9–10] [1, Sec. 1.1.3, pp. 12–14].

1.1.2 Blood as a Continuum

Consider a time-dependent family of spatial fluid domains. For a final time $T_{\text{final}} > 0$, use $I_T := (0, T_{\text{final}})$ for interior-time PDE statements and $\overline{I}_T := [0, T_{\text{final}}]$ for initial data and flow maps. Define

$$\begin{cases} \Omega_B(t) := \{\boldsymbol{x} \in \mathbb{R}^3 : \boldsymbol{x} \text{ lies inside the vessel at time } t\} \\ \mathcal{Q}_T := \{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}, \end{cases}$$

where the reference configuration is denoted $\Omega_B(0)$, the current configuration is $\Omega_B(t)$, and the space-time fluid domain is $\mathcal{Q}_T$. This domain notation is the one collected in Appendix B; boundary and function-space conventions are summarized in Appendix D. In this report, the continuum assumption is a large-vessel idealization: individual blood cells, platelets, and plasma constituents are not resolved, and blood velocity, pressure, density, and Cauchy stress are represented by sufficiently regular fields on $\mathcal{Q}_T$. Cellular-scale effects may enter only through later constitutive choices [2, Part I, Ch. 1, Secs. 1.1–1.2, pp. 4–5] [1, Secs. 1.1.3 and 3.2, pp. 12–14 and 46–47]. The continuum setup is as follows: Remark 1.1, Definition 1.2, Definitions 1.4–1.5, Definitions 1.7–1.8, and Definitions 1.12–1.13 fix the assumptions used in the model hierarchy.

1.1.3 Blood Dynamics

This subsection fixes the kinematic vocabulary used by the later continuum and reduced-order models. For $t \in \overline{I}_T$, let $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ be an orientation-preserving diffeomorphism with $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t > 0$. As a standing assumption, the family $\mathcal{L}_t$ has the time and space regularity needed to differentiate $\nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ in the classical identities below and in Appendix E. It carries each material label $\boldsymbol{\xi}$ to its current location $\boldsymbol{x}(t, \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field $\boldsymbol{u} : \mathcal{Q}_T \to \mathbb{R}^3$ satisfies

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

Figure 1 illustrates this Lagrangian–Eulerian notation.

Remark 1.1 (Eulerian vs. Lagrangian). The Eulerian description views fields such as $\phi(t, \boldsymbol{x})$ or $\boldsymbol{u}(t, \boldsymbol{x})$ at fixed spatial locations. The Lagrangian description follows material particles $\boldsymbol{\xi}$ along trajectories generated by the velocity field and views fields as functions of the material label $\boldsymbol{\xi}$ and time t, such as $\widehat{\phi}(t, \boldsymbol{\xi}) = \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$ or $\widehat{\boldsymbol{u}}(t, \boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The flow map $\mathcal{L}_t$ is the change of variables between these descriptions, and the

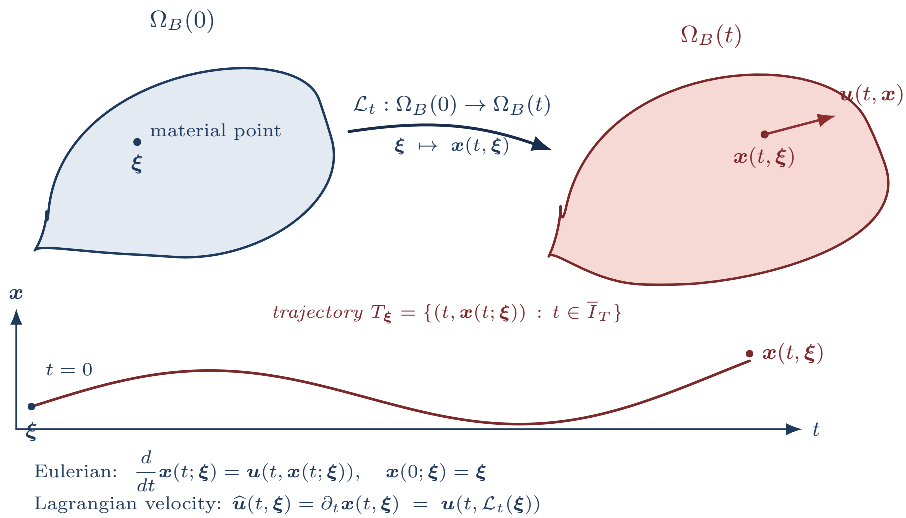

Figure 1: Lagrangian and Eulerian descriptions. The flow map $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ carries the material label $\boldsymbol{\xi} \in \Omega_B(0)$ to its current position $\boldsymbol{x}(t,\boldsymbol{\xi}) \in \Omega_B(t)$. The Eulerian velocity $\boldsymbol{u}(t,\boldsymbol{x})$ is the velocity of whichever particle currently sits at $(t,\boldsymbol{x})$; the Lagrangian velocity is $\widehat{\boldsymbol{u}}(t,\boldsymbol{\xi}) = \partial_t \boldsymbol{x}(t,\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$, and trajectories $T_{\boldsymbol{\xi}}$ are generated by the velocity field $\boldsymbol{u}$.

trajectory of a particle with label $\boldsymbol{\xi}$ is the characteristic curve $t \mapsto \boldsymbol{x}(t;\boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. Where defined, the inverse $\mathcal{L}_t^{-1}$ maps an Eulerian location back to its material label.

Under the standard assumption that the prescribed velocity field is continuous in time and uniformly Lipschitz in space, particle trajectories exist uniquely and depend continuously on their initial labels; see [3, Ch. 2]. This is a trajectory-level ODE result and should not be confused with Navier–Stokes well-posedness.

Definition 1.2 (Material derivative)**.** For $\phi \in C^1(\mathcal{Q}_T)$, the material derivative is the derivative of ϕ along a particle trajectory:

$$D_t\phi(t,\boldsymbol{x}) := \frac{d}{dt}\phi(t, \mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \boldsymbol{u}(t,\boldsymbol{x}) \cdot \nabla\phi(t,\boldsymbol{x}), \qquad \boldsymbol{\xi} = \mathcal{L}_t^{-1}(\boldsymbol{x}).$$

For vector fields, D_t is applied componentwise.

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ be the deformation gradient and $\widehat{J}_t = \det \widehat{\boldsymbol{F}}_t$ the Jacobian determinant. In Eulerian notation this is written $J_t(\boldsymbol{x}) = \det \widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$. The standard Jacobian evolution law $D_t J_t = J_t \operatorname{div} \boldsymbol{u}$ is used implicitly below; derivation details are deferred to Appendix E.

Theorem 1.3 (Reynolds Transport Theorem)**.** *Let $V(t) = \mathcal{L}_t(V(0))$ be a bounded material volume with Lipschitz boundary. If $\phi \in C^1(\mathcal{Q}_T)$, then*

$$\begin{aligned}\frac{d}{dt}\int_{V(t)} \phi \, d\boldsymbol{x} &= \int_{V(t)} (D_t\phi + \phi \, div\boldsymbol{u}) \, d\boldsymbol{x} \\ &= \int_{V(t)} (\partial_t\phi + div(\phi\boldsymbol{u})) \, d\boldsymbol{x}.\end{aligned}$$

Proof-level details for Theorem 1.3 and the momentum balance below are found in Appendix E.

Continuum assumptions retained in the main model

Let the 3D velocity, pressure, and density fields be

$$\boldsymbol{u} : \mathcal{Q}_T \to \mathbb{R}^3, \qquad p : \mathcal{Q}_T \to \mathbb{R}, \qquad \rho : \mathcal{Q}_T \to \mathbb{R}^+.$$

The balance-law vocabulary developed below permits variable density. The reduced arterial models considered later adopt the constant-density specialization $\rho \equiv \rho_0 > 0$, which is standard for large-vessel incompressible blood-flow models.

The domains and material control volumes are assumed regular enough for traces, outward unit normals, and the divergence theorem; Lipschitz boundaries are sufficient for the integral manipulations used here; see Convention D.1 in Appendix D.

Definition 1.4 (Material density preservation)**.** A flow has material density preservation if density is preserved along particle trajectories:

$$D_t \rho = 0.$$

Definition 1.5 (Incompressible flow)**.** A flow is incompressible, or volume preserving, if material volumes are preserved. With $\mathcal{L}_0 = \mathrm{Id}$ this is equivalent to

$$J_t \equiv 1 \quad \Longleftrightarrow \quad \mathrm{div}\boldsymbol{u} = 0.$$

Remark 1.6 (Divergence-free condition)*.* Mass conservation gives

$$\partial_t \rho + \mathrm{div}(\rho \boldsymbol{u}) = 0 \quad \Longleftrightarrow \quad D_t \rho + \rho \mathrm{div}\boldsymbol{u} = 0.$$

In the constant-density model used for incompressible Navier–Stokes, $\rho \equiv \rho_0 > 0$, so mass conservation reduces to $\mathrm{div}\boldsymbol{u} = 0$ in $\mathcal{Q}_T$.

Definition 1.7 (Linear momentum balance)**.** For a material fluid element $V(t) \subset \Omega_B(t)$, the linear momentum balance is

$$\frac{d}{dt} \int_{V(t)} \rho \boldsymbol{u} \, dV = \int_{\partial V(t)} \boldsymbol{T} \hat{\boldsymbol{n}} \, dS + \int_{V(t)} \rho \boldsymbol{f_b} \, dV,$$

where $\boldsymbol{T}$ is the Cauchy stress tensor and $\boldsymbol{f_b}$ is body force per unit mass.

Definition 1.8 (Strong momentum balance)**.** Under the regularity and mass-balance assumptions above, the local momentum balance is

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}, \qquad \text{in } \Omega_B(t).$$

The stress law closes the momentum equation. Let the rate-of-deformation tensor be

$$\boldsymbol{D}(\boldsymbol{u}) \;:=\; \frac{1}{2}\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\top}\right).$$

Definition 1.9. A fluid is Newtonian if, after separating the pressure contribution, its viscous stress depends linearly on the rate-of-deformation tensor $\boldsymbol{D}(\boldsymbol{u})$.

Remark 1.10 (Blood isotropy justification)*.* Blood is often modeled as an isotropic continuum fluid at large-vessel scales. This is a closure assumption for averaged behavior, not a microscopic statement about cells suspended in plasma. The cited sources support the scale and rheology caveat–nearly constant large-vessel relative viscosity versus diameter- and hematocrit-dependent apparent viscosity in small tubes–not a direct cellular-level isotropy claim [1, Sec. 1.1.3, Fig. 1.14, pp. 12–14, Sec. 3.2, pp. 46–47] [4, Abstract, pp. H1770–H1772].

Remark 1.11 (Isotropic constitutive response)*.* An isotropic fluid has a constitutive response that is independent of the chosen coordinate system. Consequently, the Cauchy stress depends on coordinate-free invariants of $\boldsymbol{D}(\boldsymbol{u})$ rather than on a particular basis.

Definition 1.12 (Newtonian, isotropic constitutive law)**.** For a Newtonian, isotropic fluid,

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}) + \lambda\,\mathrm{div}(\boldsymbol{u})\,\mathbf{I},$$

where η is the dynamic viscosity and λ is the coefficient of the volumetric-viscous term.

Definition 1.13 (Incompressible stress tensor)**.** When $\mathrm{div}\boldsymbol{u} = 0$, the Newtonian incompressible Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}).$$

More general models allow viscosity to vary with shear rate. For generalized Newtonian blood models, the dynamic viscosity is allowed to depend on the local deformation-rate magnitude [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19]. A standard constitutive form is

$$\dot{\gamma} \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})},$$

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta_{\text{eff}}(\dot{\gamma})\boldsymbol{D}(\boldsymbol{u}).$$

Here

$$\|\boldsymbol{D}(\boldsymbol{u})\|^2 \;:=\; \boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u}) = \sum_{i,j} D_{ij}(\boldsymbol{u})^2$$

is the pointwise Frobenius norm squared. When a source writes the viscosity as a function of $D : D$ or $|D|_{\mathrm{F}}^2$, this report translates it through the scalar shear-rate convention above. Here η_{eff} is an effective dynamic viscosity. Kinematic viscosity is $\nu = \eta/\rho$ after division by density. The Newtonian incompressible model is recovered from this generalized Newtonian form when η_{eff} is constant.

Remark 1.14 (Newtonian Blood Justification). For large arteries, empirical studies suggest that apparent viscosity is often approximately constant across the relevant operating range, motivating the Newtonian approximation in many reduced-order hemodynamic models. However, diameter-dependent, hematocrit-dependent, and shear-dependent effects become increasingly important in smaller vessels and low-shear regions [1, Sec. 1.1.3, pp. 12–14] [4].

1.2 Governing Equations and Model Hierarchy

The continuum assumptions of Section 1.1 lead naturally to a hierarchy of mathematical models used throughout computational hemodynamics. The hierarchy ranges from fully resolved three-dimensional descriptions to highly reduced lumped-parameter models.

1.2.1 Conservation Laws and Navier–Stokes Specialization

The governing equations originate from the conservation of mass and linear momentum applied to an incompressible continuum fluid; the derivation details and row-divergence convention are recorded in Appendix E and Definition D.6.

Mass conservation requires that fluid volume be preserved, yielding

$$\operatorname{div}\boldsymbol{u} = 0 \qquad \text{in } \Omega_B(t).$$

Linear momentum conservation balances inertial, pressure, viscous, and body forces,

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) \;=\; \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b},$$

where ρ denotes density, $\boldsymbol{T}$ is the Cauchy stress tensor, and $\boldsymbol{f_b}$ represents body forces per unit mass [2, Part I, Ch. 1] [5].

Substituting the incompressible Newtonian stress law from Definition 1.13, and assuming constant dynamic viscosity, gives the incompressible Navier–Stokes specialization

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \eta\Delta\boldsymbol{u} + \rho\boldsymbol{f_b},$$
$$\mathrm{div}\boldsymbol{u} = 0.$$

The corresponding generalized-Newtonian incompressible model keeps the same kinematic constraint but replaces the constant-viscosity diffusion term by a shear-rate-dependent viscous stress:

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mathbf{div}(2\eta_{\text{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})) + \rho\boldsymbol{f_b},$$
$$\mathrm{div}\boldsymbol{u} = 0.$$

Together with boundary and initial data, these conservation principles form the foundation of the cardiovascular flow models considered throughout this report.

1.2.2 Boundary and Initial Data Vocabulary

For a single-vessel three-dimensional domain, write the boundary decomposition

$$\partial\Omega_B(t) = \Gamma_{\text{in}}(t) \cup \Gamma_{\text{out}}(t) \cup \Gamma_w(t).$$

Here $\Gamma_{\text{in}}(t)$ denotes an inlet portion where a velocity, flow-rate, or profile condition may be prescribed. The outlet portion $\Gamma_{\text{out}}(t)$ carries pressure, traction, or reduced downstream network data. The wall portion $\Gamma_w(t)$ carries either a no-slip or no-penetration condition for a fixed-wall continuum model, or a wall-law/structure closure when wall motion is retained.

1.2.3 Model Hierarchy

Model level	State variables	Domain	Main assumptions	Role in this report
3D Navier–Stokes	$(\boldsymbol{u}, p)$	$\boldsymbol{x} \in \Omega_B(t) \subset \mathbb{R}^3$	Continuum, incompressible, Newtonian or generalized Newtonian	Reference continuum model
2D/axisymmetric	Reduced velocity and pressure fields	Meridional or cross-sectional domain	Symmetry or geometric simplification of the 3D setting	Intermediate model class
1D area-flow	(A, Q, p_{g})	$z \in (0, L)$	Cross-sectional averaging, wall law, and friction closure	Numerical-study target
0D/lumped	Pressures and flows	Graph or network	Spatially lumped compartments and algebraic/differential closure laws	Boundary and network context

A Acronyms and Abbreviations

General abbreviations.

a.e.	almost everywhere	bpm	beats per minute
e.g.	*exempli gratia* (for example)	i.e.	*id est* (that is)

Clinical, modeling, and numerical acronyms.

CAS	coronary artery stenosis	FSI	fluid–structure interaction
CCTA	coronary computed tomography angiography	RBC	red blood cell
CT-FFR	computed tomography-derived fractional flow reserve	FFR	fractional flow reserve
IVUS	intravascular ultrasound	WSS	wall shear stress
OCT	optical coherence tomography	CFD	computational fluid dynamics
ODE	ordinary differential equation	PDE	partial differential equation
IC	initial condition	BC	boundary condition
FDM	finite difference method	FEM	finite element method
FVM	finite volume method	ROM	reduced-order model
0D	zero-dimensional	1D	one-dimensional
2D	two-dimensional	3D	three-dimensional

B Mathematical Notation

Notation B.1 (Report mathematical notation)**.** This appendix collects the mathematical notation used throughout this report.

Symbol	**Meaning**
$:=$	defines
$\equiv$	notational equivalence / identified with
$\mathbb{R}$	real numbers
$\mathbb{R}^+$	strictly positive real numbers
$\mathbb{R}^-$	strictly negative real numbers
$\mathbb{R}^n$	n-dimensional real vector space
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	nonnegative integers $\{0, 1, 2, \ldots\}$
$\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}\}$	general basis of $\mathbb{R}^n$
$\{\boldsymbol{e_1}, \ldots, \boldsymbol{e_n}\}$	standard basis of $\mathbb{R}^n$
$[n]$	index set $\{1, 2, \ldots, n\} \subset \mathbb{N}$ for $n \in \mathbb{N}$
I_T	open time interval $(0, T_{\text{final}})$ for transient interior-time arguments
$\overline{I}_T$	closed time interval $[0, T_{\text{final}}]$ for initial-time data and flow maps
K	compact space-time rectangle, such as $[0, \ell] \times [0, T_{\text{final}}]$, used for output-field norms when locally stated
$\Omega \subset \mathbb{R}^n$	bounded domain
$\overline{\Omega}$	the closure of Ω
$\partial\Omega$	the boundary of Ω
$\Omega_B(t)$	spatial blood domain at time t
$\mathcal{Q}_T$	space-time fluid region $\{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}$
$C^k(\Omega)$	functions with continuous classical derivatives through order k in Ω
$C^k(\overline{\Omega})$	functions whose derivatives through order k extend continuously to $\overline{\Omega}$
$C_c^k(\Omega)$	k times continuously differentiable functions with compact support in Ω
$C_c^\infty(\Omega)$	smooth functions with compact support in Ω

Symbol	Meaning
$C(K)$	continuous functions on a stated compact set K
$L^p(\Omega)$	Lebesgue space of p-integrable functions modulo equality a.e.
$d\boldsymbol{x}$	Lebesgue volume element on $\mathbb{R}^n$
$dS_{\boldsymbol{x}}$	surface measure element on $\partial\Omega \subset \mathbb{R}^n$
dV	3D volume element on domain $\Omega \subset \mathbb{R}^3$
dS	3D surface element on boundary $\partial\Omega \subset \mathbb{R}^3$
∇	gradient operator
$\Delta\phi = \nabla \cdot \nabla\phi$	Laplacian of scalar field ϕ
div	divergence of a vector field
div	row-wise divergence of a tensor field
$\boldsymbol{v}_i$	i-th component of vector $\boldsymbol{v}$
$\boldsymbol{S} : \boldsymbol{T}$	Frobenius contraction $\sum_{i,j} S_{ij}T_{ij}$ for second-order tensors
$\langle\cdot,\ \cdot\rangle_X$	inner product on vector space X
$\langle\boldsymbol{u},\ \boldsymbol{v}\rangle \ \equiv\ \langle\boldsymbol{u},\ \boldsymbol{v}\rangle_{\mathbb{R}^n}$	inner product of vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
$\partial_{\hat{\boldsymbol{n}}}\phi$	normal derivative $\langle\nabla\phi,\ \hat{\boldsymbol{n}}\rangle$ of scalar field ϕ on $\partial\Omega$
$\lVert\cdot\rVert_X$	norm on a specified normed space X; bare $\lVert\cdot\rVert$ is local shorthand
$\lVert\cdot\rVert_{\mathrm{F}}$	Frobenius norm of a matrix or second-order tensor

C Domain Specific Notation

Definition C.1 (SI unit convention)**.** Unless otherwise stated, physical units are SI: length in meters, time in seconds, density in $\mathrm{kg/m^3}$, pressure in Pa, dynamic viscosity in $\mathrm{Pa \cdot s}$, and kinematic viscosity in $\mathrm{m^2/s}$.

Convention C.2 (Pressure-reference and ratio convention)**.** Pressure-ratio outputs are reported only with a recorded pressure reference p_{ref} and a gauge pressure p_{g} in the same reference frame. The reference-consistent pressure used in ratios is

$$p_{\mathrm{ratio}} \;:=\; p_{\mathrm{ref}} + p_{\mathrm{g}}.$$

The ratio-output specification is

$$\pi_{\mathrm{PR}} \;:=\; (z_{\mathrm{prox}}, z_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}}).$$

The subscript HF labels a declared high-flow averaging window in the study record; it is not a clinical hyperemia model unless an external measurement comparison is separately specified [6, Ch. 5, Sec. 5.5, pp. 72–73] [7, p. 1703]. The ratio-output specification is admissible for a computed pressure trace when $\mathcal{T}_{\mathrm{HF}} \subset I_T$ is Lebesgue-measurable with $0 < |\mathcal{T}_{\mathrm{HF}}| < \infty$, the proximal and distal p_{ratio} traces are integrable on that window, and

$$\langle p_{\mathrm{ratio}}(z_{\mathrm{prox}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}} \neq 0.$$

For an admissible π_{PR}, the corresponding nonclinical pressure ratio is

$$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}}) \;:=\; \frac{\langle p_{\mathrm{ratio}}(z_{\mathrm{dist}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}{\langle p_{\mathrm{ratio}}(z_{\mathrm{prox}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}.$$

For an integrable scalar trace,

$$\langle f\rangle_{\mathcal{T}_{\mathrm{HF}}} \;:=\; \frac{1}{|\mathcal{T}_{\mathrm{HF}}|}\int_{\mathcal{T}_{\mathrm{HF}}} f(t)\,dt.$$

This bracket denotes time averaging, not the inner product notation in Appendix B. An FFR-like shorthand refers to this reduced ratio only for records carrying an admissible π_{PR}; the shorthand does not add a clinical measurement model [6, Ch. 5, Sec. 5.5, pp. 72–73] [7, p. 1703]. If π_{PR} is absent or inadmissible, the pressure ratio is outside the reported output space.

Symbol	Meaning	Units
R	vessel radius; vessel diameter is $d_v := 2R$	m
d_v	vessel diameter	m
η	dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu := \eta/\rho$	kinematic viscosity	m^2/s
$\eta_{\text{eff}}(\dot{\gamma})$	effective dynamic viscosity for generalized Newtonian models	$\mathrm{Pa} \cdot \mathrm{s}$
τ	shear stress	Pa
$\dot{\gamma}$	shear rate	s^{-1}
ρ	density field	kg/m^3
p	pressure field	Pa
$\boldsymbol{u}$	velocity field	m/s
$D(\boldsymbol{u})$	rate-of-deformation tensor, $D(\boldsymbol{u}) := \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top\right)$	s^{-1}
$W_0 := d_v \sqrt{\frac{\omega\rho}{\eta}}$	Womersley number, diameter-based convention	$-$
Re	Reynolds number	$-$
Pe	Péclet number	$-$
c_{conc}	concentration field when a transport scalar is discussed	$-$
D_m	molecular diffusion coefficient	m^2/s
t	time	s
T or T_{final}	terminal time; use T_{final} where it could be confused with stress tensors	s
ω	angular frequency	rad/s
(r, θ, z)	cylindrical coordinates used only when 3D coordinate forms are written	$\mathrm{m}, -, \mathrm{m}$
$(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$	cylindrical orthonormal basis vectors	$-$
τ_{ij}	cylindrical viscous-stress component	Pa
$\boldsymbol{f_b}$	body force per unit mass	m/s^2
$\rho \boldsymbol{f_b}$	body force per unit volume	N/m^3

Reduced-model and forward-map notation.

Symbol	Meaning	Units
$A(z,t)$	cross-sectional lumen area in the 1D model	m^2
$Q(z,t)$	volumetric flow rate through a cross-section	m^3/s
$\bar{u}(z,t) := Q/A$	section-mean axial velocity	m/s
$I_z = (0, L)$	axial single-vessel interval for a 1D formulation	m
$\mathbf{U} = (A, Q)^\top$	conservative area-flow state vector	$(\mathrm{m}^2, \mathrm{m}^3/\mathrm{s})$
α	prescribed momentum-flux coefficient in the 1D balance-law closure; the baseline flat-profile convention is $\alpha = 1$	–
$R(A(z,t)) = \sqrt{A(z,t)/\pi}$	current reduced lumen radius derived from the 1D area	m
$R_0(z)$	reference stenosed radius used by the wall law	m
$R_{\mathrm{base}}(z)$	nominal healthy baseline radius before applying $s(z)$	m
$\mathcal{R}$	rheology descriptor in a_{1D}; distinct from radius symbols R, R_0, and R_{base}	mixed
$s(z)$	dimensionless stenosis profile, with $0 \leq s(z) < 1$	–
$A_0(z) = \pi R_0(z)^2$	wall-law reference area after applying the stenosis profile	m^2
$s_{\max}$	maximum fractional radius-reduction parameter in $s(z)$	–
z_c	preferred notation for center of the stenosis profile	m
z_s	alias for center of the stenosis profile used in legacy figures or prior notation	m
ℓ_s	preferred notation for length of the stenosis profile	m
L_s	alias for length of the stenosis profile used in legacy figures or prior notation	m
$\beta(z)$	wall-stiffness parameter in $p_{\mathrm{g}} = p_{\mathrm{ext}} + \beta A_0^{-1}(\sqrt{A} - \sqrt{A_0})$	$\mathrm{Pa} \cdot \mathrm{m}$
$p_{\mathrm{ext}}(z,t)$	external-pressure datum in the wall-law gauge convention; baseline value $p_{\mathrm{ext}} \equiv 0$ unless declared otherwise	Pa
$\Psi(A,z,t)$	elastic flux potential satisfying $\partial_A \Psi = (A/\rho)\partial_A p_{\mathrm{g}}$ and normalized by $\Psi(A_0(z),z,t) = 0$	$\mathrm{m}^4/\mathrm{s}^2$
$\mathbf{F}$	area-flow flux map; not a residual or objective functional	problem-dependent
$\mathbf{S}$	source-term map in the 1D balance-law template	problem-dependent

Symbol	Meaning	Units
$\widehat{\mathbf{F}}^n_{i+1/2}$	numerical flux at a finite-volume cell interface and time level	problem-dependent
$\widehat{\mathbf{S}}^n_i$	cell source quadrature used by a computed finite-volume scheme	problem-dependent
$\Delta z_i, \Delta t^n$	finite-volume cell length and timestep in a computed solve	m, s
C_f or $C_f(z, A)$	wall-friction coefficient in the 1D source $-C_f Q/A$; first-run value $C_f = 8\pi\eta/\rho$	m^2/s
$\eta_{\mathrm{fric}}(z, t)$	dynamic viscosity used by the reduced wall-friction and signed-shear-proxy formula; $\eta_{\mathrm{fric}} = \eta$ for constant-viscosity records and $\eta_{\mathrm{fric}} = \eta_{\mathrm{eff}}(\dot{\gamma}_w)$ for the admitted Carreau–Yasuda wall-friction extension	$\mathrm{Pa} \cdot \mathrm{s}$
c_{wave} or c	1D pulse-wave speed, $c^2 = (A/\rho)\partial_A p_{\mathrm{g}}$ when no concentration field is in scope	$\mathrm{m/s}$
$e \in \mathcal{E}$	expert/closure record for a closure-indexed 1D forward-map family; suppress e only for single-expert baseline records	mixed
$\mathcal{G}^{r_h, e}_{1D,h}$	computed fixed-grid 1D map indexed by numerical record r_h and expert record e	mixed

Symbol	Meaning	Units
p_{3D}	continuum pressure in the 3D/NS setting	Pa
$p_{3D,\mathrm{g}}$	continuum pressure after subtracting the wall-law gauge reference	Pa
p_{g}	1D gauge pressure produced by the wall law	Pa
$p_{\mathrm{ratio}} := p_{\mathrm{ref}} + p_{\mathrm{g}}$	reference-consistent pressure used in ratios by Convention C.2	Pa
p_{ref}	fixed pressure reference recorded by the ratio-output specification	Pa
$\pi_{\Delta p} = (z_{\mathrm{prox}}, z_{\mathrm{dist}})$	pressure-drop tap specification for the base 1D output map	m, m
π_{PR}	ratio-output specification $(z_{\mathrm{prox}}, z_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}})$	mixed
$\langle f \rangle_{\mathcal{T}_{\mathrm{HF}}}$	time average of an integrable trace over the declared high-flow window; not the Appendix B inner product	same as f
$p_{\mathrm{out,g}}$	prescribed outlet gauge pressure in the same reference frame as p_{g}	Pa
$\Delta p(t)$	proximal-minus-distal pressure drop computed from p_{g}	Pa
$\overline{\Delta p}_{\mathcal{T}_{\mathrm{HF}}}$	scalar pressure-drop output averaged over recorded high-flow window $\mathcal{T}_{\mathrm{HF}}$	Pa
FFR-like shorthand	optional shorthand for PR_{1D} only when Convention C.2 and admissibility requirements are met	–
$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$	schematic nonclinical pressure-ratio quantity under the same convention	–
τ_w or $\lvert\tau_w\rvert$	resolved traction-based WSS magnitude in 3D contexts	Pa
$\tau_{w,z}^{1D}$	signed 1D axial shear proxy from Q, A, and the recorded wall-friction/rheology closure	Pa

Remark C.3 (Viscosity notation). We write η for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. The distinction is dimensional: this report follows the convention that the coefficient in the Cauchy stress tensor is a dynamic viscosity: $D(\boldsymbol{u})$ has units of inverse time, so $\eta D(\boldsymbol{u})$ has pressure units. For generalized Newtonian models, the scalar viscosity function multiplying $2D(\boldsymbol{u})$ in the extra stress is likewise an effective dynamic viscosity, denoted

$$\eta_{\mathrm{eff}}(\dot\gamma), \qquad \dot\gamma \ := \ \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

Here “this quantity” means the stress-level viscosity coefficient, not the divided-by-density coefficient. Several sources write the generalized viscosity function as $\mu(|D|^2)$, while related hemodynamics sources also use μ for viscosity coefficients in reduced and continuum settings [1, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [8, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18]. To keep this report unambiguous, ν is always kinematic viscosity, while η and η_{eff} are dynamic viscosities.

D Mathematical Foundations

This appendix supplies citation anchors for the analytic notation used in the main chapters and in Appendix B. It separates domain and boundary language, classical function spaces, field notation, differential operators, and norm conventions so later sections can cite only the convention they need.

Convention D.1 (Standing regularity convention)**.** Unless a section states a stronger or weaker hypothesis, classical identities are used only under the regularity needed for their terms to be defined. In particular, fields have the displayed classical derivatives in the indicated domains, boundary traces are defined when boundary terms are written, and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated explicitly; Sobolev traces are not inferred from this classical convention. At minimum, boundaries are assumed piecewise C^1 when classical boundary integrals or integration-by-parts identities are used. Outward unit normals are then interpreted on the regular boundary pieces, hence surface-a.e. When a pointwise normal derivative is asserted on all of $\partial\Omega$, the boundary is assumed at least C^1.

Definition D.2 (Domain and boundary notation)**.** Let $\Omega \subset \mathbb{R}^n$ be a bounded open connected set, with $n \in \mathbb{N}$, and denote its closure by $\overline{\Omega}$. In the hemodynamics settings considered in this report, only $n \in [3]$ is used. The standing regularity convention in Convention D.1 applies whenever traces, outward unit normals, or integration by parts are used. For open Ω,

$$\overline{\Omega} = \Omega \cup \partial\Omega, \qquad \partial\Omega = \overline{\Omega} \setminus \Omega.$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded and is therefore compact by the Heine–Borel theorem. The open set Ω itself need not be compact. Since Ω is bounded, $\partial\Omega$ is also compact. When the standing regularity convention supplies a piecewise C^1 boundary, the outward unit normal $\hat{\boldsymbol{n}}$ is interpreted on the regular boundary pieces, hence $dS_{\boldsymbol{x}}$-a.e. on $\partial\Omega$.

Definition D.3 (Time intervals and compact space-time sets)**.** For a final time $T_{\text{final}} > 0$, this report uses

$$I_T \;:=\; (0, T_{\text{final}}), \qquad \overline{I}_T \;:=\; [0, T_{\text{final}}].$$

The open interval I_T is used for interior-time differential identities and space-time PDE regions, while $\overline{I}_T$ is used when initial-time data, endpoint data, sampled time grids, or flow maps are included. When a local final time such as T_* is stated, the same convention applies with T_{final} replaced by that local final time.

For one-dimensional output fields, a compact space-time rectangle is written as

$$K \;:=\; [0, \ell] \times [0, T_{\text{final}}]$$

or by an explicitly stated local variant. Spaces such as $C(K)$ and $C^1([0,\ell])$ mean continuous functions on the named compact set, with the supremum norm when a norm is written. Spaces such as $L^p(K)$ use the product Lebesgue measure and the same almost-everywhere convention as Definition D.7.

Definition D.4 (Field and multi-index notation)**.** For $\Omega \subset \mathbb{R}^n$, we use the following field notation:

$$\begin{aligned} \phi : \Omega \to \mathbb{R} \quad & \text{is a } \textit{scalar field}, \\ \boldsymbol{f} : \Omega \to \mathbb{R}^n \quad & \text{is a } \textit{vector field}, \\ \boldsymbol{T} : \Omega \to \mathbb{R}^{n\times n} \quad & \text{is a } \textit{(second-order) tensor field}, \end{aligned}$$

and we write $\phi(\boldsymbol{x})$, $\boldsymbol{f}(\boldsymbol{x})$, and $\boldsymbol{T}(\boldsymbol{x})$[1] for the values at a point $\boldsymbol{x} \in \Omega$. When referring to a physical quantity, its measurement units are indicated as $\left[\ \cdot\ \right]$, whereas dimensionless quantities are indicated as $\left[\ -\ \right]$. The k-th partial derivative of ϕ with respect to coordinate x_i is

$$\partial_{x_i}^k \phi \;\equiv\; \frac{\partial^k \phi}{\partial x_i^k}.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$, set $|\alpha| = \alpha_1 + \cdots + \alpha_n$. When the derivative exists classically, the corresponding derivative of a scalar field ϕ is

$$D^\alpha \phi \;\equiv\; \frac{\partial^{|\alpha|}\phi}{\partial x_1^{\alpha_1}\, \partial x_2^{\alpha_2}\, \ldots\, \partial x_n^{\alpha_n}}.$$

For $|\alpha| = 1$, $D^\alpha \phi$ is a first-order partial derivative of ϕ, e.g. $\partial_{x_i}\phi$ where $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \in [n] \setminus \{i\}$.

Definition D.5 (Classical function-space notation)**.** Given a domain Ω and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, a scalar function space on Ω is an $\mathbb{F}$-vector subspace of the maps $f : \Omega \to \mathbb{F}$. Unless a complex-valued space is explicitly specified, this report uses real-valued spaces. Thus $C^0(\Omega)$ denotes the real-valued continuous functions on the open domain Ω. For $k \in \mathbb{N}_0$, $C^k(\Omega)$ denotes functions $\phi : \Omega \to \mathbb{R}$ such that $D^\alpha\phi$ exists and is continuous in Ω for every $|\alpha| \le k$. By contrast, $C^k(\overline{\Omega})$ denotes functions in $C^k(\Omega)$ whose derivatives through order k extend continuously to the closure $\overline{\Omega}$. The space $C^\infty(\Omega)$ is the intersection of all $C^k(\Omega)$ for $k \in \mathbb{N}_0$, i.e. the infinitely differentiable functions on Ω. The notation $C^k(\Omega; \mathbb{R}^m)$ denotes the maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$ with components in $C^k(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in C^k(\Omega)$ for all $i \in [m]$.

Definition D.6 (Differential-operator conventions)**.** For real-valued $\phi \in C^1(\Omega)$, the partial derivative with respect to coordinate x_i is

$$\partial_{x_i}\phi = \frac{\partial \phi}{\partial x_i}.$$

[1] A second-order tensor value $\boldsymbol{T}(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ can be identified with the matrix of a linear map $\mathbb{R}^n \to \mathbb{R}^n$ acting on vectors by matrix-vector multiplication.

For $\boldsymbol{x} \in \Omega$ and a direction $\boldsymbol{v} \in \mathbb{R}^n$, the directional derivative is

$$D_{\boldsymbol{v}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \boldsymbol{v}\rangle_{\mathbb{R}^n}.$$

If $\phi \in C^1(\overline{\Omega})$, $\boldsymbol{x} \in \partial\Omega$, and $\hat{\boldsymbol{n}}$ is the outward unit normal at $\boldsymbol{x}$, then the boundary normal derivative is

$$\partial_{\hat{\boldsymbol{n}}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \hat{\boldsymbol{n}}\rangle_{\mathbb{R}^n}.$$

For Lebesgue measure $d\boldsymbol{x}$ on $\mathbb{R}^n$ the integral of ϕ in Ω is

$$\int_{\Omega} \phi(\boldsymbol{x})\, d\boldsymbol{x}.$$

If $\partial\Omega$ is of class C^1 and $\phi \in C^0(\overline{\Omega})$, then the classical boundary trace $\phi|_{\partial\Omega}$ is continuous on $\partial\Omega$. Since Ω is bounded and $\partial\Omega$ is a compact C^1 hypersurface, this trace belongs to $L^1(\partial\Omega;\, dS_{\boldsymbol{x}})$. The corresponding boundary integral is written

$$\int_{\partial\Omega} \phi(\boldsymbol{x})\; dS_{\boldsymbol{x}}.$$

We use the column-gradient convention for scalar fields. For $\phi \in C^1(\Omega)$, the gradient is

$$\nabla\phi \;:=\; (\partial_{x_i}\phi)_{i\in[n]} \in \mathbb{R}^n.$$

For $\phi \in C^2(\Omega)$, the Laplacian is

$$\Delta\phi \;:=\; \nabla\cdot\nabla\phi \;=\; \sum_{i\in[n]} \partial^2_{x_i}\phi.$$

For vector-valued $\boldsymbol{f} \in C^1(\Omega;\mathbb{R}^n)$ with components $\boldsymbol{f} = (f_i)_{i\in[n]}$, the vector gradient is the Jacobian with entries $(\nabla\boldsymbol{f})_{ij} \;:=\; \partial_{x_j} f_i$. The divergence is the trace of this Jacobian:

$$\operatorname{div}\boldsymbol{f} \;:=\; \nabla\cdot\boldsymbol{f} \;=\; \sum_{i\in[n]} \partial_{x_i} f_i \;=\; \operatorname{tr}(\nabla\boldsymbol{f}).$$

For $\boldsymbol{f} \in C^2(\Omega;\mathbb{R}^n)$, the Laplacian is defined componentwise. The domain integral is also componentwise whenever the components are integrable:

$$\begin{aligned}
&\Delta\,\boldsymbol{f} \;:=\; (\Delta f_i)_{i\in[n]} \\
&\int_{\Omega} \boldsymbol{f}(\boldsymbol{x})\; d\boldsymbol{x} \;:=\; \left(\int_{\Omega} f_i(\boldsymbol{x})\; d\boldsymbol{x}\right)_{i\in[n]}.
\end{aligned}$$

For tensor-valued $\boldsymbol{T} \in C^1(\Omega;\mathbb{R}^{n\times n})$ with entries $\boldsymbol{T} = (T_{ij})_{i,j\in[n]}$, we use the row-divergence convention $(\mathbf{div}\boldsymbol{T})_i = \sum_{j\in[n]} \partial_{x_j} T_{ij}$. Equivalently, if $\boldsymbol{t}_i = (T_{ij})_{j\in[n]}$ denotes the i-th row of $\boldsymbol{T}$, then

$$
\begin{aligned}
\mathbf{div}\,\boldsymbol{T} &:= (\operatorname{div}\boldsymbol{t}_i)_{i\in[n]} \\
&= (\nabla\cdot\boldsymbol{t}_i)_{i\in[n]} \\
&= \left(\sum_{j\in[n]} \partial_{x_j} T_{ij}\right)_{i\in[n]}.
\end{aligned}
$$

For $\boldsymbol{T} \in C^2(\Omega;\mathbb{R}^{n\times n})$, the Laplacian is defined componentwise. The domain integral is also defined componentwise whenever the entries are integrable:

$$
\begin{aligned}
\Delta\boldsymbol{T} &:= (\Delta T_{ij})_{i,j\in[n]} \\
\int_\Omega \boldsymbol{T}(\boldsymbol{x})\; d\boldsymbol{x} &:= \left(\int_\Omega T_{ij}(\boldsymbol{x})\; d\boldsymbol{x}\right)_{i,j\in[n]}.
\end{aligned}
$$

Definition D.7 (C^k and L^p norm conventions)**.** Let $\Omega \subset \mathbb{R}^n$ be bounded and open. We write $C^k(\overline{\Omega})$ for the functions $\phi \in C^k(\Omega)$ such that each derivative $D^\alpha\phi$, $|\alpha| \le k$, extends continuously to $\overline{\Omega}$. The norm is

$$
\|\phi\|_{C^k(\overline{\Omega})} := \sum_{|\alpha|\le k} \sup_{\boldsymbol{x}\in\overline{\Omega}} |D^\alpha\phi(\boldsymbol{x})|.
$$

With this norm, $C^k(\overline{\Omega})$ is a Banach space: every Cauchy sequence in $C^k(\overline{\Omega})$ converges, with respect to $\|\cdot\|_{C^k(\overline{\Omega})}$, to a limit in $C^k(\overline{\Omega})$. In particular, for $k = 0$, we have

$$
\|\phi\|_{C^0(\overline{\Omega})} := \sup_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})| = \max_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})|\,.
$$

For continuous functions on compact sets, the max norm agrees with the usual supremum norm.

For $1 \le p < \infty$, $L^p(\Omega)$ denotes the space of equivalence classes of real-valued measurable functions on Ω, identified up to equality almost everywhere, with finite norm

$$
\|\phi\|_{L^p(\Omega)} := \left(\int_\Omega |\phi(\boldsymbol{x})|^p\, d\boldsymbol{x}\right)^{1/p}.
$$

For $p = \infty$,

$$
\|\phi\|_{L^\infty(\Omega)} := \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} |\phi(\boldsymbol{x})|.
$$

With this almost-everywhere identification, $L^p(\Omega)$ is a Banach space for $1 \le p \le \infty$.

The notation $L^p(\Omega;\mathbb{R}^m)$ denotes equivalence classes of measurable maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$, identified up to equality a.e., whose components belong to $L^p(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in L^p(\Omega)$ for all $i \in [m]$. When a norm on $L^p(\Omega;\mathbb{R}^m)$ is needed, we use the Euclidean norm in the range. For $1 \le p < \infty$,

$$\|\boldsymbol{f}\|_{L^p(\Omega;\mathbb{R}^m)} \;:=\; \left(\int_\Omega \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}^p \, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$, the corresponding convention is

$$\|\boldsymbol{f}\|_{L^\infty(\Omega;\mathbb{R}^m)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}.$$

For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm

$$\|\boldsymbol{T}\|_{\mathrm{F}} \;:=\; (\boldsymbol{T} : \boldsymbol{T})^{1/2} = \left(\sum_{i,j\in[n]} T_{ij}^2\right)^{1/2}.$$

Thus tensor-valued L^p norms are formed by replacing $|\phi(\boldsymbol{x})|$ in the scalar L^p formula with $\|\boldsymbol{T}(\boldsymbol{x})\|_{\mathrm{F}}$.

E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the first controlled 1D study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. The standing regularity convention, boundary notation, and differential-operator convention are those of Convention D.1, Definition D.2, and Definition D.6. In particular, **div** is the row-divergence of a tensor field, and boundary forms assume the stated trace and normal hypotheses; no weak formulation is being asserted in this appendix.

E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t, \boldsymbol{\xi}) := \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t, \mathcal{L}_t(\boldsymbol{\xi})) = \partial_t \phi(t, \boldsymbol{x}) + \nabla\phi(t, \boldsymbol{x}) \cdot \boldsymbol{u}(t, \boldsymbol{x}), \qquad \boldsymbol{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\mathrm{div}(\phi \boldsymbol{u}) = \nabla\phi \cdot \boldsymbol{u} + \phi \mathrm{div} \boldsymbol{u}$$

also gives

$$D_t \phi = \partial_t \phi + \mathrm{div}(\phi \boldsymbol{u}) - \phi \mathrm{div} \boldsymbol{u}.$$

E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\widehat{J}_t = \det \widehat{\boldsymbol{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving C^1 change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\boldsymbol{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t \widehat{J}_t = \widehat{J}_t \, \mathrm{tr}\Big(\partial_t \widehat{\boldsymbol{F}}_t \, \widehat{\boldsymbol{F}}_t^{-1}\Big) = \widehat{J}_t \, \mathrm{div} \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_t J_t = J_t \mathrm{div} \boldsymbol{u}$ [2, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \boldsymbol{x})\, d\boldsymbol{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\widehat{J}_t(\boldsymbol{\xi})\, d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} (D_t\phi + \phi \text{div}\boldsymbol{u})\ d\boldsymbol{x} = \int_{V(t)} (\partial_t\phi + \text{div}(\phi\boldsymbol{u}))\ d\boldsymbol{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi\boldsymbol{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t\phi\ d\boldsymbol{x} + \int_{\partial V(t)} \phi\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dS.$$

E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho\boldsymbol{f_b}\, dV,$$

Reynolds transport gives

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{V(t)} (D_t(\rho\boldsymbol{u}) + \rho\boldsymbol{u}\, \text{div}\boldsymbol{u})\, dV.$$

Using mass conservation, $D_t\rho + \rho\text{div}\boldsymbol{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t\boldsymbol{u}\, dV = \int_{V(t)} \rho\, (\partial_t\boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u})\, dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS = \int_{V(t)} \mathbf{div}(\boldsymbol{T})\, dV.$$

This is the tensor row-divergence convention from Definition D.6, applied under the regularity needed for the stress trace $\boldsymbol{T}\hat{\boldsymbol{n}}$ and the volume integral of $\mathbf{div}\boldsymbol{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho\, (\partial_t\boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}.$$

The conservative Eulerian form

$$\partial_t(\rho \boldsymbol{u}) + \mathbf{div}(\rho \boldsymbol{u} \otimes \boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho \boldsymbol{u}) + \mathbf{div}(\rho \boldsymbol{u} \otimes \boldsymbol{u}) = \rho \left(\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}\right) + \boldsymbol{u} \left(\partial_t \rho + \operatorname{div}(\rho \boldsymbol{u})\right).$$

F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \boldsymbol{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and η is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.3.

F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity η and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \mathbf{div}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top) = \eta \Delta \boldsymbol{u} + \eta \nabla(\mathrm{div} \boldsymbol{u}).$$

For incompressible flow, $\mathrm{div} \boldsymbol{u} = 0$, so

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}, \qquad \rho^{-1} \mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \nu \Delta \boldsymbol{u}, \qquad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory. This context is separate from the verification evidence required for the reduced 1D solver used later.

F.2 Cylindrical Coordinate Form

For (r, θ, z) with orthonormal basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad \boldsymbol{u} = u_r \boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$
\begin{aligned}
(\boldsymbol{u}\cdot\nabla) &= u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z,\\
\nabla p &= \partial_r p\,\boldsymbol{e}_r + \frac{1}{r}\partial_\theta p\,\boldsymbol{e}_\theta + \partial_z p\,\boldsymbol{e}_z,\\
\mathrm{div}\boldsymbol{u} &= \frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z.
\end{aligned}
$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$
\begin{aligned}
&D_{rr} = \partial_r u_r, && D_{\theta\theta} = \frac{1}{r}\partial_\theta u_\theta + \frac{u_r}{r}, && D_{zz} = \partial_z u_z,\\
&D_{r\theta} = \frac{1}{2}\left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r}\partial_\theta u_r\right), && D_{rz} = \frac{1}{2}(\partial_r u_z + \partial_z u_r), && D_{\theta z} = \frac{1}{2}\left(\frac{1}{r}\partial_\theta u_z + \partial_z u_\theta\right).
\end{aligned}
$$

Use the scalar shear-rate convention $\dot\gamma = \sqrt{2\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{u})}$. Then

$$
\tau_{ij} = 2\eta_{\mathrm{eff}}(\dot\gamma)D_{ij}, \qquad i,j\in\{r,\theta,z\}.
$$

When a source writes the law as a function of $D:D$ or $|D|_{\mathrm{F}}^2$, this report translates it through this convention, consistent with Remark C.3. Writing $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_z\boldsymbol{e}_z$, the momentum components are

$$
\begin{aligned}
\text{(radial)}\quad & \partial_t u_r + u_r\partial_r u_r + \frac{u_\theta}{r}\partial_\theta u_r + u_z\partial_z u_r - \frac{u_\theta^2}{r}\\
&\quad = -\frac{1}{\rho}\partial_r p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rr}) + \frac{1}{r}\partial_\theta\tau_{r\theta} + \partial_z\tau_{rz} - \frac{1}{r}\tau_{\theta\theta}\right] + f_r,\\
\text{(azimuthal)}\quad & \partial_t u_\theta + u_r\partial_r u_\theta + \frac{u_\theta}{r}\partial_\theta u_\theta + u_z\partial_z u_\theta + \frac{u_r u_\theta}{r}\\
&\quad = -\frac{1}{\rho r}\partial_\theta p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{r\theta}) + \frac{1}{r}\partial_\theta\tau_{\theta\theta} + \partial_z\tau_{\theta z} + \frac{1}{r}\tau_{r\theta}\right] + f_\theta,\\
\text{(axial)}\quad & \partial_t u_z + u_r\partial_r u_z + \frac{u_\theta}{r}\partial_\theta u_z + u_z\partial_z u_z\\
&\quad = -\frac{1}{\rho}\partial_z p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rz}) + \frac{1}{r}\partial_\theta\tau_{\theta z} + \partial_z\tau_{zz}\right] + f_z.
\end{aligned}
$$

The incompressibility condition is

$$
\frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z = 0.
$$

G Code Availability and Build Statement

This report is built from the local LaTeX sources rooted at `notes.tex`. The compiled PDF should be treated as the rendered form of those sources. Any numerical-study code, solver configuration, or generated data used later in the report should be cited in the relevant study section with enough information to identify the method, inputs, and output records.

References

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